GW matching: the composite formulation

Mathematical formulation of the two methods used in the experiment

Let $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{m\times m}$ be symmetric distance matrices for two views of a dataset. Let $a \in \mathbb{R}^n_{++}$ and $b \in \mathbb{R}^m_{++}$ sum to one; our experiments use uniform weights. The variable $P \in \mathbb{R}^{n\times m}$ is a transport coupling. Define

$$f(P) = \sum_{i,k=1}^{n}\sum_{j,\ell=1}^{m}(A_{ik}-B_{j\ell})^2 P_{ij}P_{k\ell},$$

$$C = \{P \geq 0 : P\mathbf{1}_m = a\}, \qquad T(P) = P^\top \mathbf{1}_n,$$

$$D = \{b\}, \qquad g(y) = \iota_D(y) = \begin{cases} 0, & y = b, \\ +\infty, & y \neq b. \end{cases}$$

Thus C imposes nonnegativity and row masses; $g \circ T$ imposes column masses. The benchmark is exactly

$$\boxed{\min_{P\in C} f(P) + g(TP)} \iff \min_{P\in U(a,b)} f(P), \qquad U(a,b) := \{P \geq 0 : P\mathbf{1}_m = a,\ P^\top \mathbf{1}_n = b\}.$$

Here f is smooth and generally nonconvex, g is a convex indicator, C is compact and convex, and T is linear. There is no entropy regularization in this objective. Matrix inner products are Frobenius inner products, so

$$T^* u = \mathbf{1}_n u^\top, \qquad \langle P, T^* u\rangle_F = \langle TP, u\rangle, \qquad \|T\| = \sqrt{n}.$$

Stochastic FraMES: smooth g, and keep the simple constraint C. For $\mu > 0$, the proximal map and Moreau envelope of g are explicit:

$$\mathrm{prox}_{\mu g}(y) = b, \qquad g_\mu(y) = \min_z \left\{ g(z) + \frac{\|z-y\|_2^2}{2\mu} \right\} = \frac{\|y-b\|_2^2}{2\mu}.$$

FraMES therefore takes Frank–Wolfe steps for the changing smooth surrogate

$$F_\mu(P) = f(P) + \frac{\|P^\top \mathbf{1}_n - b\|_2^2}{2\mu}, \qquad \nabla(g_\mu \circ T)(P) = \frac{\mathbf{1}_n (P^\top \mathbf{1}_n - b)^\top}{\mu}.$$

Let $\widehat{G}_k$ be a fresh unbiased estimate of $\nabla f(P_k)$. Momentum is applied to this estimate, while the column-penalty gradient is evaluated at the current iterate:

$$\begin{aligned} &d_0 = \widehat{G}_0, \qquad d_k = (1-\rho_k)d_{k-1} + \rho_k \widehat{G}_k \quad (k \geq 1), \\ &H_k = d_k + \mu_k^{-1} T^*(TP_k - b), \\ &S_k \in \arg\min_{S\in C} \langle H_k, S\rangle_F, \qquad P_{k+1} = (1-\gamma_k)P_k + \gamma_k S_k. \end{aligned}$$

The linear minimization oracle separates by row: choose $j_i \in \arg\min_j (H_k)_{ij}$, set $(S_k)_{ij_i} = a_i$, and set the other entries to zero. Each step preserves $P_k \in C$; column masses may differ from b until repaired or approached through decreasing smoothing. The implemented schedules are

$$\gamma_k = \gamma_0 (k+1)^{-3/4}, \qquad \rho_k = \rho_0 (k+1)^{-1/2}, \qquad \mu_k = \frac{s_\mu}{m}(k+1)^{-1/4}.$$

To match the manuscript's $T = \mathrm{Id}$ notation, set $\widetilde{D} = \{P : P^\top \mathbf{1}_n = b\}$. Then $\mathrm{dist}_F(P, \widetilde{D})^2 = \|TP - b\|_2^2/n$; its smoothing parameter $\beta = \mu/n$ gives precisely the same penalty. The gradient of the penalty above has Lipschitz constant n/μ.

The same problem with entropic mirror descent

Stochastic entropic mirror descent: enforce C and $TP = b$ in each proximal step. Absorb the composite constraint into $X = U(a, b) = C \cap T^{-1}(D)$. In the notation of Fatkhullin–He (2024), take the smooth expected objective to be f, the convex regularizer to be zero, and the distance-generating function to be negative entropy:

$$\omega(P) = \sum_{ij} P_{ij}(\log P_{ij} - 1), \qquad D_\omega(Q, P) = D_{\mathrm{KL}}(Q\|P) = \sum_{ij}\left[Q_{ij}\log\frac{Q_{ij}}{P_{ij}} - Q_{ij} + P_{ij}\right].$$

With a positive initial coupling and a fresh gradient estimate $\widehat{G}_k$, its ideal update is

$$\begin{aligned} P_{k+1} &= \operatorname*{arg\,min}_{Q\in C}\left\{\langle \widehat{G}_k, Q\rangle_F + g(TQ) + \eta_k^{-1} D_{\mathrm{KL}}(Q\|P_k)\right\} \\ &= \boxed{\operatorname*{arg\,min}_{Q\in U(a,b)}\left\{\langle \widehat{G}_k, Q\rangle_F + \eta_k^{-1} D_{\mathrm{KL}}(Q\|P_k)\right\}.} \end{aligned}$$

Thus $g\circ T$ remains an exact constraint in the mathematical subproblem. There is no Moreau smoothing or momentum in this method. Its entropy term defines the update geometry; the objective being minimized remains the same f.

The KL subproblem is solved by matrix scaling. With componentwise operations, let

$$K_k = P_k \odot \exp(-\eta_k \widehat{G}_k), \qquad P_{k+1} = \mathrm{diag}(u) K_k \,\mathrm{diag}(v).$$

Sinkhorn alternates $u \leftarrow a/(K_k v)$ and $v \leftarrow b/(K_k^\top u)$ to enforce both marginals. This is a *KL/Bregman projection* of K_k onto $U(a, b)$, rather than a Euclidean projection. Our implementation uses stable log-domain arithmetic and accepts a numerical step only when both the summed marginal ℓ_1 residual and normalized guarded primal–dual gap are at most 10^{-7}.

For $C_{\max} = \max_{i,k,j,\ell}(A_{ik} - B_{j\ell})^2 > 0$, the implemented admissible family is

$$\eta_k = \frac{c}{4C_{\max}\sqrt{k+1}}, \qquad 0 < c \le 1.$$

Indeed, on $U(a, b)$ the quadratic remainder is bounded by $C_{\max}\|Q - P\|_1^2 \le 2C_{\max} D_{\mathrm{KL}}(Q\|P)$, yielding a relative-smoothness constant $2C_{\max}$. The paper's update assumes an exact proximal minimizer; our finite inner tolerance is a numerical choice, with no inexact outer theorem claimed.

The common stochastic oracle. Both methods sample the gradient of the full, fixed GW objective. For $r = P\mathbf{1}_m$, $c_P = P^\top \mathbf{1}_n$, and entrywise squares $A^{\odot 2}, B^{\odot 2}$,

$$\nabla f(P) = 2(A^{\odot 2} r)\mathbf{1}_m^\top + 2\mathbf{1}_n (B^{\odot 2} c_P)^\top - 4APB.$$

The current marginals must be used here: FraMES iterates need not satisfy $c_P = b$. For fresh $Z_k \in \{-1, 1\}^{m\times q}$ with independent entries and $q = 128$, the implementation uses

$$\widehat{G}_k = 2(A^{\odot 2} r)\mathbf{1}_m^\top + 2\mathbf{1}_n (B^{\odot 2} c_P)^\top - \frac{2}{q}\left[(AP_kBZ_k)Z_k^\top + (AP_kZ_k)(BZ_k)^\top\right].$$

Since $\mathbb{E}[Z_k Z_k^\top/q] = I_m$, this is conditionally unbiased: $\mathbb{E}[\widehat{G}_k \mid P_k] = \nabla f(P_k)$. FraMES averages these estimates through d_k; SMD uses $\widehat{G}_k$ directly. Both use one oracle call per update; FraMES reuses the first call for d_0.

The computational distinction. FraMES handles the column constraint through its explicit quadratic penalty and uses only a rowwise linear minimization oracle. SMD solves a KL transport projection imposing both marginals each update. Reported endpoint losses evaluate the same f after a common transport-repair procedure. Those empirical losses are separate from the methods' respective stationarity guarantees and SMD's prescribed randomly selected output.

Sources: the supplied *Nonsmooth_FW-7.pdf*, Proposition 2.11 and Corollary 2.10; and Fatkhullin–He, *Taming Nonconvex Stochastic Mirror Descent with General Bregman Divergence*, AISTATS 2024, Eq. (2), Thm. 4.3. Formulas checked against campaign source `38b5ea8`.